METRIC
FF-B-171/14
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING, ANSI/AFBMA DIMENSION SERIES 23.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

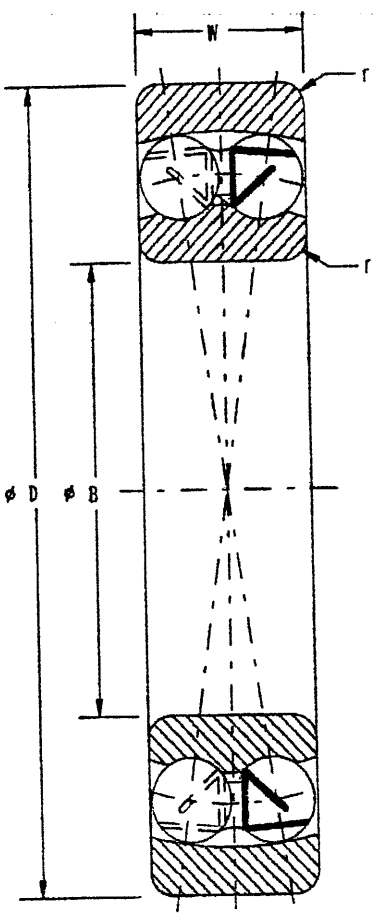


FIGURE 1.
DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING WITH CYLINDRICAL BORE (CODE C)

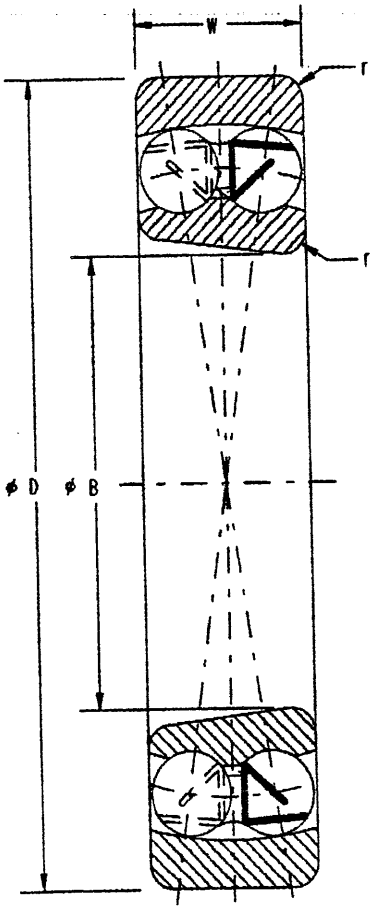


FIGURE 2.
DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING WITH TAPERED BORE (CODE T)

BEARING CONFIGURATIONS

FSC 3110

AMSC/NA

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Design. All self-aligning bearings shall be open. Bores shall be cylindrical or tapered (taper 1:12) and shall be specified in the part number by the codes in table III.

4. Misalignment. Maximum permissible misalignment shall be 4.0 degrees.

5. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

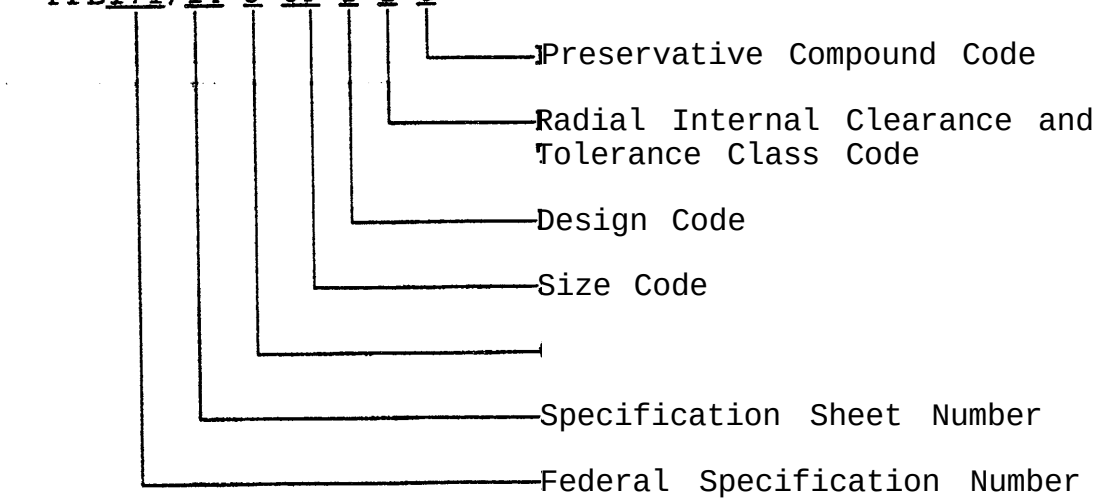
6. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

7. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code “C” in the part number.

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8. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, bore design, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE: FFB171/14 J 09 T 1 C



FFB171/14J09T1C indicates - Bearing, cage-pressed steel or molded Plastic; bore 45mm; outside diameter 100mm; width 36mm; open (no shields or seals), tapered bore; radial internal clearance symbol 2 (less than normal); tolerance class ABEC-1; preservative compound in accordance with MIL-P-197.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	35	17	0.6	15	1750
01	12	37	17	1.0	18	1985
02	15	42	17	1.0	21	2060
03	17	47	19	1.0	23	2500
04	20	52	21	1.1	27	3085
05	25	62	24	1.1	32	4135
06	30	72	27	1.1	37	5300
07	35	80	31	1.5	43.5	6635
08	40	90	33	1.5	48.5	7590
09	45	100	36	1.5	53.5	9280
10	50	110	40	2.0	60	11070
11	55	120	43	2.0	65	12780
12	60	130	46	2.1	72	14865
13	65	140	48	2.1	77	16058
14	70	150	51	2.1	82	17625
15	75	160	55	2.1	87	19380
16	80	170	58	2.1	92	21910
17	85	180	60	3.0	99	23665
18	90	190	64	3.0	104	25420
19	95	200	67	3.0	109	27920
20	100	215	73	3.0	114	31140
21	105	225	77	3.0	119	32995
22	110	240	80	3.0	124	35055

1/ Size codes in Table I are industry bore codes.
2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.
3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

TABLE III. Design Codes.

Code	Design
C	Open (no shields or seals), Cylindrical bore - fig.1
T	Open (no shields or seals), Tapered bore - fig.2

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

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